When Does Text Help? A Leakage-Free Information-Richness and Encoder-Adaptation Study for Multimodal Land-Cover Composition Regression

Ömer Faruk Merey
Graduate School of Informatics, Middle East Technical University
omer.merey@metu.edu.tr

Abstract—We study whether auxiliary text captions improve land-cover composition regression from satellite imagery on ARAS400k, under strict no-leakage conditions. The dataset's hybrid captions embed the ground-truth percentages verbatim, so we restrict text to the leakage-free vision-only captions and add numeric sanitization. We evaluate an information-richness axis (empty caption, no-text floor, keywords, two qualitative captions, random text) with frozen CLIP encoders, then introduce the project's principal ablation: unfreezing the encoder via LoRA. Three results emerge. (1) With frozen encoders, the cross-attention block (not the text) drives the gain, and clean captions help only when they correctly describe the image (a fidelity-conditional effect that averages near zero). (2) LoRA fine-tuning lowers test MSE by 32–60 % and reverses the architectural story: once the encoder adapts, the pure-vision model becomes the best and the fusion layer no longer helps. (3) A controlled leakage condition (raw hybrid captions) collapses MSE to ∼1/5 of the best leakage-free model, quantifying the failure mode that motivated the redesign.

Index Terms—remote sensing, land cover, multimodal learning, CLIP, cross-attention, LoRA, data leakage, ablation

I. Introduction

Land-cover composition is a structured regression target: seven mutually exclusive classes (Tree, Shrub, Grass, Crop, Built-up, Barren, Water) summing to 100 %. ARAS400k [6] provides 10 000 RGB tiles, segmentation masks, and five caption variants generated by vision-language models (VLMs). The project asks whether captions improve composition regression beyond a vision-only baseline, and if so by how much.

The original design was found to contain a structural data-leakage path: the hybrid_* and text_qwen3-4b captions state the ground-truth percentages verbatim (e.g. "grasslands (92 %)"). A model trained against them solves the task lexically rather than multimodally. Crucially, the leak is not merely lexical: the VLMs that wrote those captions saw the percentages, so even stripping the digits leaves a label-shaped narrative. We therefore restrict text to the two vision_* captions (generated from the image alone) and apply numeric sanitization as a safety filter.

Research question. How does the semantic richness of auxiliary text captions, from a vision-only baseline, to keywords, to detailed qualitative descriptions, impact land-cover composition regression while strictly avoiding data leakage, and how does this change when the visual encoder is allowed to adapt?

Contributions. (1) A leakage-free information-richness ablation with frozen encoders, plus a random-text control and a caption-fidelity stratified evaluation. (2) A LoRA encoder-adaptation ablation that severely changes the results and reverses the frozen-regime conclusion about the fusion layer. (3) A controlled leakage condition that quantifies, rather than merely asserts, the leakage failure mode.

II. Related Work

CLIP [1] aligns image and text via contrastive pre-training; its adaptation to remote sensing is studied in RemoteCLIP [2] and RS5M [3], which report that domain adaptation of the encoder is the dominant lever for downstream gains, consistent with our LoRA findings. Cross-attention fusion has been shown to outperform late fusion in remote-sensing segmentation [4]; we find that this advantage is contingent on the encoder being frozen. Parameter-efficient fine-tuning via low-rank adapters [5] lets us adapt a frozen CLIP cheaply, which is what turns the architectural story around in Section V.

III. Methodology

A. Architecture

A frozen CLIP ViT-B/32 encodes the image into 49 patch tokens and the caption into 77 token embeddings. An 8-head cross-attention block uses image patches as queries and text tokens as keys/values; the 49 fused vectors are mean-pooled and passed to a Linear(512→256)→ReLU→Linear(256→7)→Softmax head. Loss is MSE on the softmax output against the target distribution. The pure-vision variant removes cross-attention (patches → mean-pool → head).

B. Conditions

The information-richness axis is realised by six frozen conditions plus a random-text control (Table I); numbers in all text are masked to a [NUM] placeholder to prevent residual leakage. R1 keywords are noun lemmas extracted with spaCy over both vision_* captions and are deliberately not filtered to the target classes, which would inject

TABLE I
Conditions. CA = cross-attention; GAP = patch mean-pool.

Code	Arch.	Text input
R0a	CLIP+CA+MLP	empty token
R0b	CLIP+GAP+MLP	— (no text)
R1	CLIP+CA+MLP	noun-lemma keywords (both vision_*)
R2a	CLIP+CA+MLP	sanitized vision_gemma3-4b
R2b	CLIP+CA+MLP	sanitized vision_qwen3-vl-8b
R-rand	CLIP+CA+MLP	77 random vocab tokens (control)
Phase 3 additions		
*-lora	+ LoRA on encoder	R0a/R0b/R2b, encoder unfrozen
Rleak	CLIP+CA+MLP	raw hybrid_gemma3-4b (GT %, leaky)

GT structure. A caption-fidelity stratified test additionally tags each test tile agree/disagree by whether a synonym of the dominant GT class appears in the conditioning caption.

C. LoRA encoder adaptation (principal ablation)

The severe change is unfreezing CLIP via low-rank adapters. We inject LoRA (r=8, α=16) into the MLP projections of the last six residual blocks of both encoders; the packed-QKV attention projection is left untouched because nn.MultiheadAttention bypasses its module forward. Only the adapters, the cross-attention block, and the head are trained ($\sim$1.4 M parameters); the CLIP base stays frozen. We fine-tune in fp32 and apply gradient clipping (max-norm 1.0), which removed a divergence observed for the empty-caption (R0a) condition, the lowest-signal case for the fusion layer. We re-train R0a, R0b, and R2b with LoRA and compare against their frozen counterparts.

D. Leakage quantification

Rleak trains on the raw, unsanitized hybrid_gemma3-4b caption (percentages intact) as a controlled negative control, a failure mode to measure, not a method to advocate.

E. Training and reproducibility

AdamW (lr 10^{-4} frozen, 5×10^{-5} LoRA, weight decay 10^{-4}), cosine annealing, early stopping, deterministic 70/15/15 split, three seeds per condition. Because frozen CLIP outputs are constant across epochs, frozen runs read precomputed fp16 embeddings (the full frozen grid runs in under an hour on an Apple M4); LoRA runs are online ($\sim$84 s/epoch on the same machine). Runs are tracked on Weights & Biases.

IV. Results

A. Frozen regime

Table II and Fig. 3 give the frozen-encoder test MSE. Removing cross-attention (R0b) is by far the largest effect; among caption-bearing conditions the empty caption (R0a) is numerically best, and the empty caption significantly beats random text (p=8.6×10^{-9}), so caption content is not irrelevant but is dominated by other factors. The fidelity-stratified test (Table III) shows why captions look unhelpful on average: R2a/R2b match or beat R0a on tiles whose caption mentions the dominant class, and underperform by a comparable margin when it does not.

TABLE II
Frozen-encoder test MSE/MAE, mean ± std over 3 seeds.

Cond.	MSE ↓	MAE ↓
R0a	.00707±.00006	.0406±.0003
R0b	.01101±.00032	.0554±.0011
R1	.00760±.00016	.0425±.0008
R2a	.00751±.00026	.0422±.0009
R2b	.00739±.00020	.0421±.0007
R-rand	.00787±.00008	.0425±.0002

TABLE III
Caption-fidelity stratified test MSE (frozen).

Cond.	MSE (agree)	MSE (disagree)
R0a	.00676	.00740
R2a	.00671	.00823
R2b	.00680	.00787

B. LoRA regime (principal ablation)

Table IV and Fig. 1 report the frozen-vs-LoRA comparison. LoRA lowers MSE by 32–60 % on every condition. The ordering flips: in the frozen regime R0a (cross-attention) far outperforms R0b (pure vision); under LoRA the pure-vision R0b becomes the best condition, and the cross-attention conditions no longer help.

C. Leakage quantification

Rleak attains test MSE 0.00143 ± 0.0001, about one third of the best leakage-free model (R0b-lora, 0.00435) and one fifth of the frozen R0a baseline (Fig. 2). The model recovers the targets almost directly from the percentages in the text.

V. Discussion

Frozen regime: architecture, not text. The dominant frozen effect is architectural, the cross-attention block, even with an empty caption, gives a $\sim$36 % lower MSE than the pure-vision floor. Without an architecture-matched no-text baseline (R0b) this would have been miscredited to the text. Clean captions provide only a fidelity-conditional benefit (Table III) that cancels on average, and noun keywords are statistically indistinguishable from random text. “Adding captions” is thus not a uniform intervention: fidelity, not richness, gates the effect.

Encoder adaptation reverses the story. The principal ablation shows that the frozen-regime conclusion is regime-dependent. Once LoRA lets the visual encoder adapt, the pure-vision model (R0b-lora, 0.00435) overtakes the cross-attention conditions (R0a-lora 0.00478, R2b-lora 0.00466): the fusion layer’s contribution in the frozen regime was effectively a capacity substitute for a non-adapted encoder, and it becomes redundant, even mildly harmful over an empty/degenerate text path, once the encoder itself can learn. This aligns with RemoteCLIP/RS5M [2], [3], where encoder domain-adaptation is the dominant lever, and

TABLE IV
Frozen vs. LoRA test MSE (mean over 3 seeds) and relative change.

Cond.	Frozen	LoRA	Δ
R0a	.00707	.00478±.0001	−32 %
R0b	.01101	.00435±.0001	−60 %
R2b	.00739	.00466±.0002	−37 %

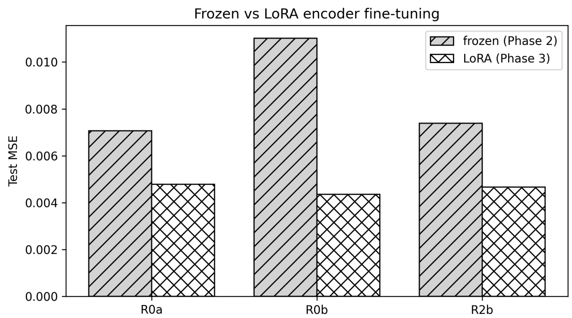

Fig. 1. Frozen vs. LoRA test MSE. LoRA improves all conditions; the pure-vision R0b benefits most and becomes the best model.

qualifies the cross-attention advantage reported for frozen fusion in [4]. The caption signal remains marginal in both regimes (R2b-lora edges R0a-lora by ∼2.5 %).

Leakage, quantified. Rleak makes the Phase-1 concern concrete: percentage-bearing captions yield an MSE 3–5× lower than any leakage-free model, an "improvement" that reflects text copying rather than multimodal reasoning. This is the quantitative justification for excluding hybrid captions throughout.

Limitations. A single synth domain; LoRA on MLP projections only (attention QKV untouched); the fidelity tag is a lexical proxy; the empty-caption LoRA condition required gradient clipping for stability.

VI. Conclusion

Under leakage-free conditions, multimodal gains on ARAS400k are dominated by the visual pathway, not the text. With frozen encoders the cross-attention fusion block carries the apparent "multimodal" improvement and clean captions help only conditionally on their fidelity to the image; with a LoRA-adapted encoder the pure-vision model is best and the fusion layer becomes redundant. A controlled leakage condition confirms that percentage-bearing captions produce large but illusory gains. Methodologically, multimodal claims must be ablated against architecture-matched no-text, random-text, and encoder-adaptation baselines before any non-trivial text contribution is asserted.

Reproducibility

Code, notebooks with outputs, and requirements.txt: https://github.com/OmerFarukMerey/remote-sensing-caption-composition. W&B: https://wandb.ai/mereyomerfaruk/di725-project.

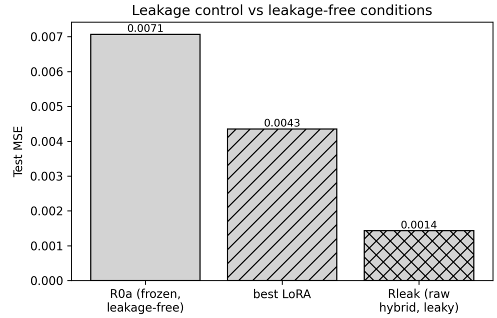

Fig. 2. Leakage control. Rleak (raw hybrid, GT % in text) collapses MSE far below the leakage-free conditions.

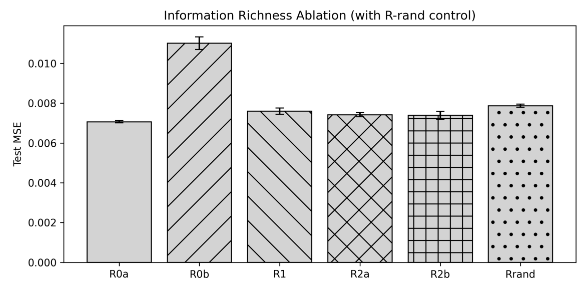

Fig. 3. Frozen-regime information-richness ablation (incl. R-rand control).

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